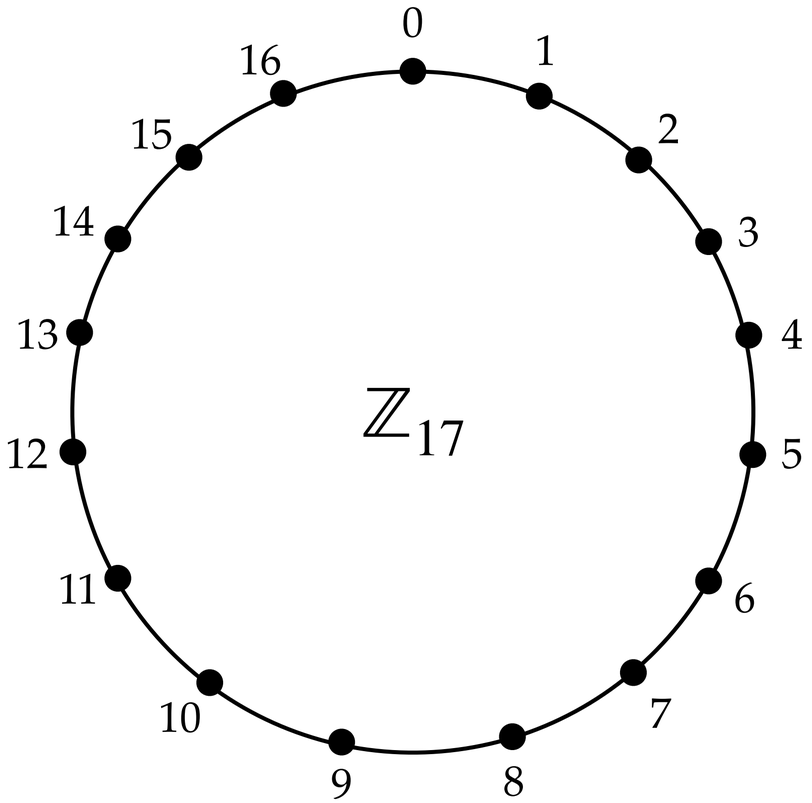
V1b: Mathematical prerequisites

1. Modular arithmetic
2. The polynomial ring
   $R_q = \mathbb{Z}_q[x]/(x^n + 1)$
3. The module R_q^k
4. "Small" polynomials
5. Lattice problems:
   MLWE, D-MLWE and MSIS
6. Why lattices?

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Modular arithmetic

- Modulus: $q \geq 2$.
- $a \equiv b \pmod{q}$ means that $a - b$ is an integer multiple of q.
- $r = a \bmod q$ means that r is the remainder upon dividing the integer a by q (so $0 \leq r \leq q-1$).
- Integers modulo q: $\mathbb{Z}_q = \{0,1,2,\ldots,q-1\}$, where addition, subtraction and multiplication are performed modulo q.
- Example: $\mathbb{Z}_{17} = \{0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16\}$.
  - In $\mathbb{Z}_{17}$, $9+15=7$, $9-15=11$, and $9 \times 15 = 16$.
  - More precisely, $9+15 = 24 \equiv 7 \pmod{17}$,
    $9-15 = -6 \equiv 11 \pmod{17}$,
    and $9 \times 15 = 135 \equiv 16 \pmod{17}$.



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Polynomial rings

- Let q be a prime modulus.
- $\mathbb{Z}_q[x]$ is the set of all polynomials in x with coefficients in $\mathbb{Z}_q$.
- When adding, subtracting, multiplying and dividing polynomials in $\mathbb{Z}_q[x]$, all coefficient arithmetic is performed in $\mathbb{Z}_q$.
- Example: Let $q = 7$, and consider $f(x) = 5 + 4x^2 + 3x^3 \in \mathbb{Z}_7[x]$ and $g(x) = 6 + 3x + 2x^2 \in \mathbb{Z}_7[x]$. Then:
  - $f(x) + g(x) = 4 + 3x + 6x^2 + 3x^3$.
  - $f(x) - g(x) = 6 + 4x + 2x^2 + 3x^3$.
  - $f(x) \times g(x) = 2 + x + 6x^2 + 2x^3 + 3x^4 + 6x^5$.

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The polynomial ring $R_q = \mathbb{Z}_q[x]/(x^n + 1)$

- Let q be a prime modulus, and let n be a positive integer.
- The polynomial ring $R_q = \mathbb{Z}_q[x]/(x^n + 1)$ is comprised of the polynomials in $\mathbb{Z}_q[x]$ of degree less than n, with multiplication of polynomials performed modulo the reduction polynomial $x^n + 1$.
- So, to multiply polynomials $f(x), g(x) \in R_q$:

  i. Multiply $f(x)$ and $g(x)$ in $\mathbb{Z}_q[x]$, obtaining a polynomial $h(x)$ of degree at most $2n - 2$.

  ii. Divide $h(x)$ by $x^n + 1$ to get a remainder polynomial $r(x)$ of degree at most $n - 1$.

  iii. Then $f(x) \times g(x) = r(x)$ in R_q.
- Note: The size of R_q is q^n.

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Example: The polynomial ring $R_q = \mathbb{Z}_{41}[x]/(x^4 + 1)$

Let $q = 41$ and $n = 4$.

- Then R_q is comprised of the polynomials in $\mathbb{Z}_{41}[x]$ of degree at most 3.
- Let $f(x) = 32 + 17x^2 + 22x^3 \in R_q$ and $g(x) = 11 + 7x + 19x^2 + x^3 \in R_q$.
- In $\mathbb{Z}_q[x]$,
  $h(x) = f(x) \times g(x) = 24 + 19x + 16x^2 + 24x^3 + 26x^4 + 25x^5 + 22x^6$.
- *The division of $h(x)$ by $x^4 + 1$ can be accomplished by replacing x^4 by -1, x^5 by $-x$, and x^6 by $-x^2$, and then simplifying.*
- We obtain $r(x) = 24 + 19x + 16x^2 + 24x^3 - 26 - 25x - 22x^2$
  $= 39 + 35x + 35x^2 + 24x^3$.
- So, $f(x) \times g(x) = 39 + 35x + 35x^2 + 24x^3$ in R_q.

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Representing polynomials as vectors

- A polynomial $f(x) = a_0 + a_1 x + \cdots + a_{n-1}x^{n-1}$ in $R_q = \mathbb{Z}_q[x]/(x^n + 1)$ can be represented by its vector of coefficients $f = (a_0, a_1, \ldots, a_{n-1})$.
  The vector has length exactly n.
- Example: Consider $R_q = \mathbb{Z}_{41}[x]/(x^4 + 1)$.
  - The polynomials $f(x) = 23 + 11x^2 + 7x^3 \in R_q$ and $g(x) = 40 + 5x + 16x^2 \in R_q$ can be represented by the vectors $f = (23, 0, 11, 7)$ and $g = (40, 5, 16, 0)$.
  - In R_q, we have $f + g = (22, 5, 27, 7)$, $f - g = (24, 36, 36, 7)$, and $f \times g = (12, 3, 29, 7)$.

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The module R_q^k

- Let k be a positive integer.
- The elements of the module R_q^k are the length-k vectors of polynomials in R_q.
- Addition and subtraction of elements in R_q^k is component-wise (so the result is also an element in R_q^k).
- The inner product (multiplication) of two vectors in R_q^k results in a polynomial in R_q.
- *All vectors in R_q^k will be written as column vectors.*

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Example: R_q^k

- Let $q = 137,\ n = 4,\ R_q = \mathbb{Z}_{137}[x]/(x^4+1),\ \ k = 3.$

- Let $a = \begin{bmatrix} 93 + 51x + 34x^2 + 54x^3 \\ 27 + 87x + 81x^2 + 6x^3 \\ 112 + 15x + 46x^2 + 122x^3 \end{bmatrix}$ and $b = \begin{bmatrix} 40 + 78x + x^2 + 119x^3 \\ 11 + 31x + 57x^2 + 90x^3 \\ 108 + 72x + 47x^2 + 14x^3 \end{bmatrix} \in R_q^k.$

- Then $a + b = \begin{bmatrix} 133 + 129x + 35x^2 + 36x^3 \\ 38 + 118x + x^2 + 96x^3 \\ 83 + 87x + 93x^2 + 136x^3 \end{bmatrix},\ a - b = \begin{bmatrix} 53 + 110x + 33x^2 + 72x^3 \\ 16 + 56x + 24x^2 + 53x^3 \\ 4 + 80x + 136x^2 + 108x^3 \end{bmatrix},$

and $a^T \cdot b = a[1]b[1] + a[2]b[2] + a[3]b[3] = 93 + 59x + 44x^2 + 132x^3.$

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Size



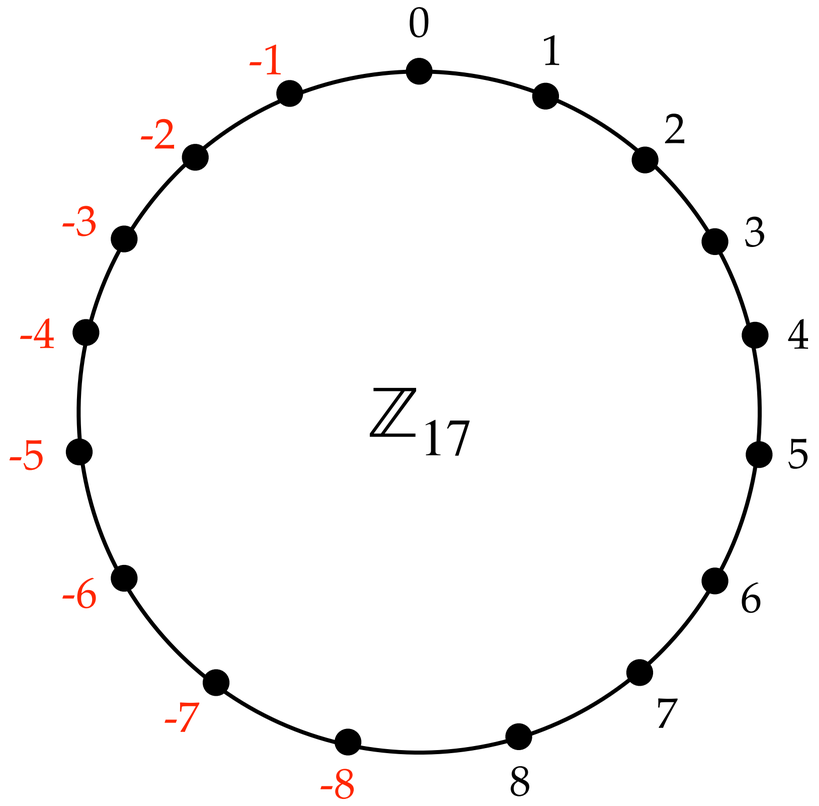
- We introduce a notion of "size" for:
  - integers in $\mathbb{Z}_q$,
  - polynomials in $R_q = \mathbb{Z}_q[x]/(x^n + 1)$, and
  - vectors of polynomials in R_q^k.
- This size is the "infinity norm", denoted by $\| \cdot \|_\infty$.
- For this, we'll need the notion of "symmetric mod".

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Symmetric mod: q odd

- Let q be odd, and $r \in \mathbb{Z}_q$.
- Then r mods $q = \begin{cases} r & \text{if } r \le (q-1)/2, \\ r-q & \text{if } r > (q-1)/2, \end{cases}$

  so $-(q-1)/2 \le r \text{ mods } q \le (q-1)/2$.
- Example: Let $q = 17$.
  - $-8 \le r \text{ mods } 17 \le 8$.
  - $6 \text{ mods } 17 = 6$, and $13 \text{ mods } 17 = -4$.
  - $9 + 15 \text{ mods } 17 = 7$.
  - $9 - 15 \text{ mods } 17 = -6$.
  - $9 \times 15 \text{ mods } 17 = -1$.
- Note: mods q is also written as $\text{mod}^{+} q$.

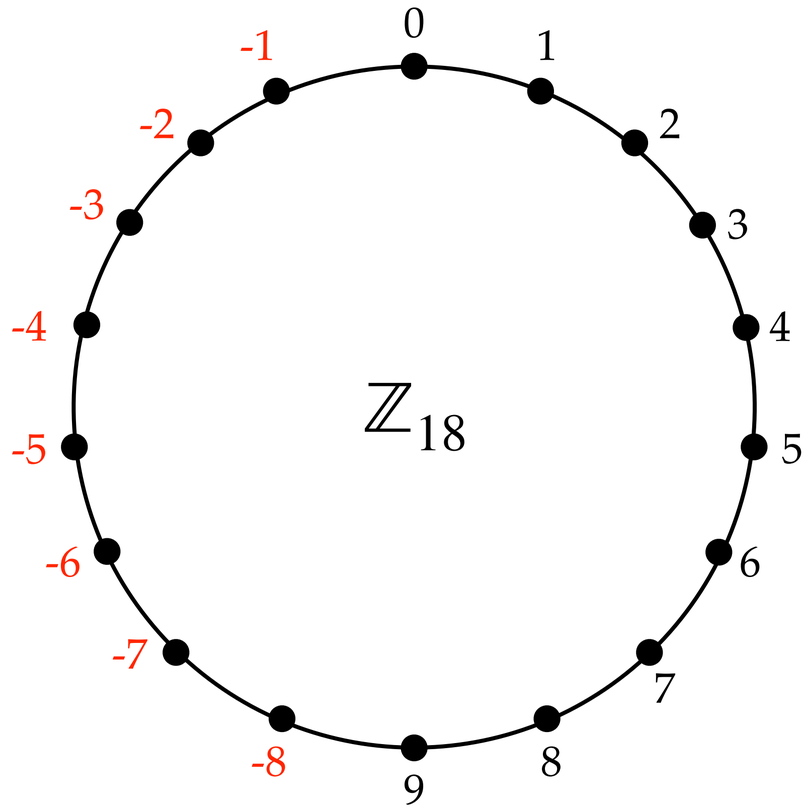


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Symmetric mod: q even

- Let q be even, and $r \in \mathbb{Z}_q$.
- Then r mods $q = \begin{cases} r & \text{if } r \le q/2, \\ r - q & \text{if } r > q/2, \end{cases}$

  so $-q/2 < r \text{ mods } q \le q/2$.
- Example: Let $q = 18$.
  - $-8 \le r \text{ mods } 18 \le 9$.
  - $6 \text{ mods } 18 = 6$, and $13 \text{ mods } 18 = -5$.
  - $9 + 15 \text{ mods } 18 = 6$.
  - $9 - 15 \text{ mods } 18 = -6$.
  - $9 \times 15 \text{ mods } 18 = 9$.
- Note: mods q is also written as $\text{mod}^{+} q$.



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Size of polynomials

- Integers modulo q. Let $r \in \mathbb{Z}_q$. Then $\|r\|_\infty = |\, r \bmod q \,|$.
  - Example: Let $q = 19$. Then $\|7\|_\infty = 7$ and $\|18\|_\infty = 1$.
  - Note that $0 \le \|r\|_\infty \le (q-1)/2$ if q is odd, and $0 \le \|r\|_\infty \le q/2$ if q is even.



- Ring elements. Let $f(x) = f_0 + f_1 x + \cdots + f_{n-1}x^{n-1} \in R_q$.
  Then $\|f\|_\infty = \max \|f_i\|_\infty$.
  - Example: Let $f(x) = 1 + 12x + 3x^3 + 18x^5 \in R_{19}$. Then $\|f\|_\infty = 7$, since the mods q representation of f is $1 - 7x + 3x^3 - x^5$.
- Module elements. Let $a = [a_1, a_2, \ldots, a_k]^T \in R_q^k$.
  Then $\|a\|_\infty = \max \|a_i\|_\infty$.

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"Small" polynomials

- A polynomial $f \in R_q$ is small if $\|f\|_\infty$ is "small".
- Let η be a positive integer that is small compared to $q/2$.
- Define $S_\eta = \{f \in R_q \mid \|f\|_\infty \leq \eta\}$ to be the set of polynomials in R_q all of whose coefficients have size at most η.
  S_η is the set of "small" polynomials.
- Example: Let $q = 31$. Then $1 + 30x + 29x^2 + x^4 + 2x^5 \in S_2$.
- Example: S_1 is the set of polynomials in R_q all of whose coefficients (when reduced mods q) are -1, 0, or 1.

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Product of small polynomials

- Claim: The product of two small polynomials in R_q is also (relatively) small.
- Example: Consider $R_q = \mathbb{Z}_{137}[x]/(x^4 + 1)$ (so $q = 137$ and $n = 4$).
  - Let $\eta = 2$, and consider $f(x) = 1 + x - 2x^2 + 2x^3 \in S_2$ and $g(x) = -2 + 2x^2 - x^3 \in S_2$.
  - Then $f(x) \cdot g(x) = 3 + 129x + 8x^2 + 134x^3 \equiv 3 - 8x + 8x^2 - 3x^3$, so $f \cdot g \in S_8$.

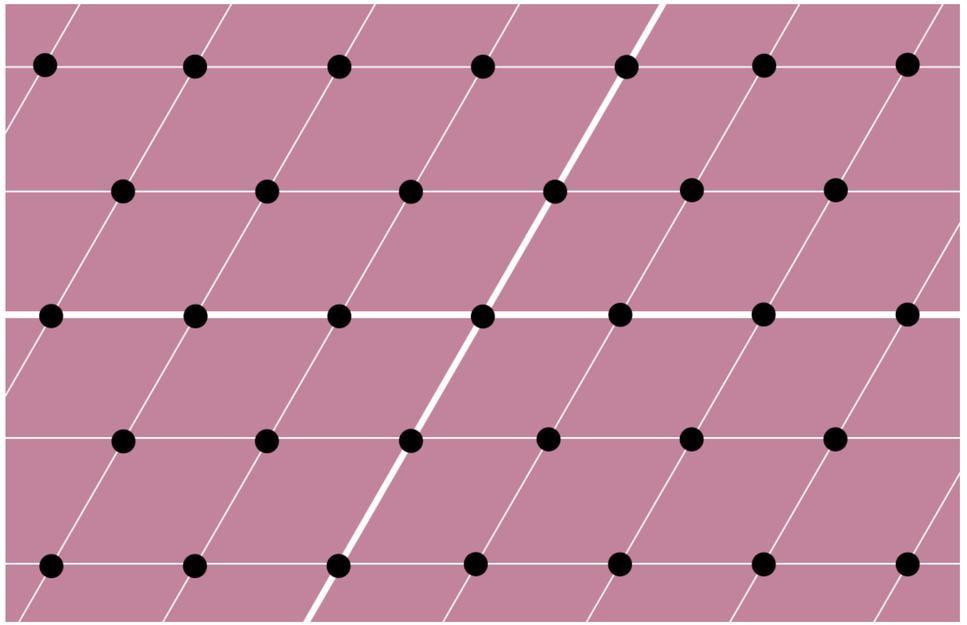
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Product of small polynomials (2)

- Claim: If $f \in S_{\eta_1}$ and $g \in S_{\eta_2}$, then $fg \in S_{n\eta_1\eta_2}$.
- Justification:
  Consider $f(x) = f_0 + f_1 x + \cdots + f_{n-1}x^{n-1} \in S_{\eta_1}$ and
  $g(x) = g_0 + g_1 x + \cdots + g_{n-1}x^{n-1} \in S_{\eta_2}$, and let
  $h(x) = f(x)g(x) = h_0 + h_1 x + \cdots + h_{n-1}x^{n-1}$.
- For each $i \in [0, n-1]$, we have
  $h_i = f_0 g_i + f_1 g_{i-1} + \cdots + f_i g_0 - f_{i+1} g_{n-1} - f_{i+2} g_{n-2} - \cdots - f_{n-2} g_{i+2} - f_{n-1} g_{i+1}$,
  so $\|h_i\|_\infty \le n\eta_1\eta_2$.
- Thus, $h \in S_{n\eta_1\eta_2}$. □
- Similarly, if $a \in S^k_{\eta_1}$ and $b \in S^k_{\eta_2}$, then $ab^T \in S_{kn\eta_1\eta_2}$.

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Lattice problems: MLWE, D-MLWE and MSIS

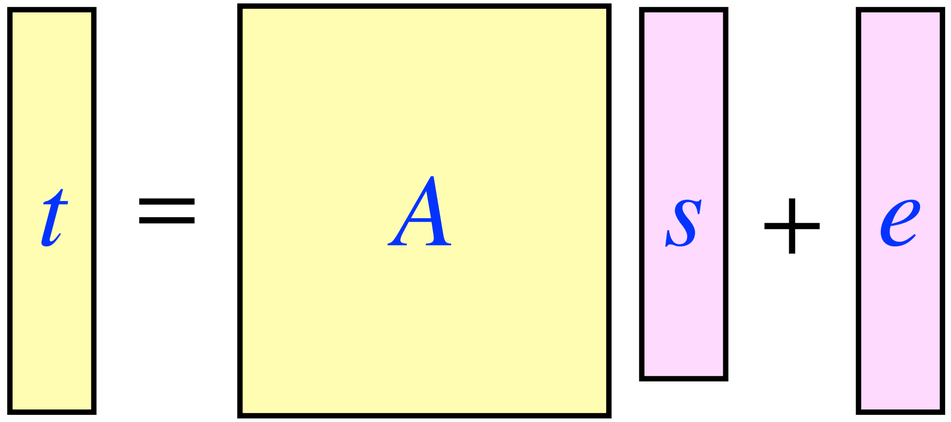


- The security of Kyber is based on the hardness of the Decisional-Module Learning With Errors (D-MLWE) problem, which in turn is related to the hardness of the MLWE problem.
- The security of Dilithium is based on the hardness of the D-MLWE and Module Short Integer Solutions (MSIS) problems.
- I'll introduce the MLWE and D-MLWE problems next.
- A description of MSIS will be deferred to the Dilithium chapter (V3).

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Lattice problem: MLWE

- Parameters:
  - prime q,
  - integers n, k, ℓ with $k \geq \ell$, and
  - integers $\eta_1, \eta_2 \ll q/2$.



- Module Learning With Errors (MLWE) instance:
  - $A \in_R R_q^{k \times \ell}$ (where $R_q = \mathbb{Z}_q[x]/(x^n + 1)$).
  - $t = As + e$, where $s \in_R S_{\eta_1}^{\ell}$ and $e \in_R S_{\eta_2}^{k}$ (so $t \in R_q^k$).
- Required: s.

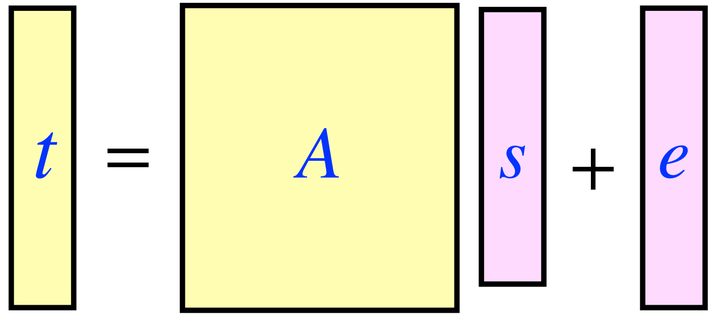
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Example: MLWE

- Parameters: $q = 541$, $n = 4$, $k = 3$, $\ell = 2$, $\eta_1 = 3$, $\eta_2 = 2$, so $R_q = \mathbb{Z}_{541}[x]/(x^4 + 1)$.
- MLWE instance generation: Randomly select

$$A = \begin{bmatrix} 442 + 502x + 513x^2 + 15x^3 & 368 + 166x + 37x^2 + 135x^3 \\ 479 + 532x + 116x^2 + 41x^3 & 12 + 139x + 385x^2 + 409x^3 \\ 29 + 394x + 503x^2 + 389x^3 & 9 + 499x + 92x^2 + 254x^3 \end{bmatrix} \in R_q^{3\times 2},$$

$$s = \begin{bmatrix} 2 - 2x + x^3 \\ 3 - 2x - 2x^2 - 2x^3 \end{bmatrix} \in S_3^2, \text{ and } e = \begin{bmatrix} 2 - 2x - x^2 \\ 1 + 2x + 2x^2 + x^3 \\ -2 - x^2 - 2x^3 \end{bmatrix} \in S_2^3,$$



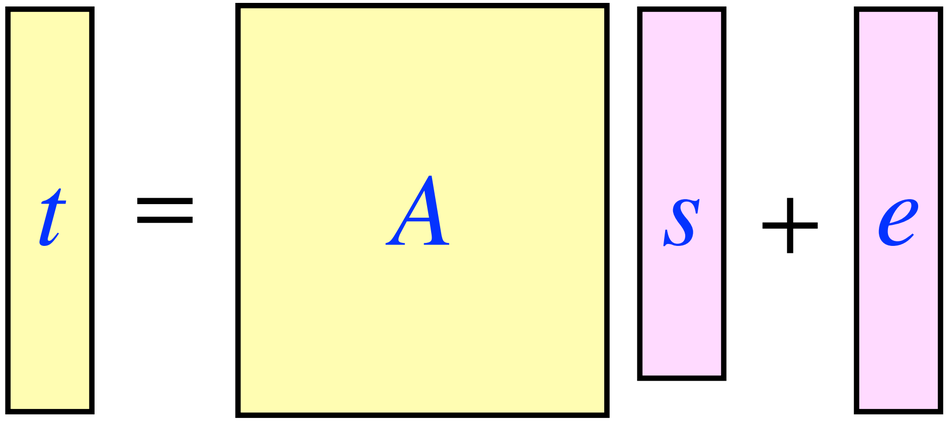
and compute $t = As + e = \begin{bmatrix} 30 + 252x + 401x^2 + 332x^3 \\ 247 + 350x + 259x^2 + 485x^3 \\ 534 + 234x + 137x^2 + 443x^3 \end{bmatrix}$. Note that $\|t\|_\infty = 259$.

- MLWE problem: Given (A, t), determine s.

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Lattice problem: D-MLWE

- Parameters:
  - prime q,
  - integers n, k, ℓ with $k \geq \ell$, and
  - integers $\eta_1, \eta_2 \ll q/2$.



  - Decisional-Module Learning With Errors (MLWE) instance:
    - $A \in_R R_q^{k \times \ell}$ (where $R_q = \mathbb{Z}_q[x]/(x^n + 1)$).
    - $z \in R_q^k$, where $z = t$ with probability $\frac{1}{2}$ (and $t = As + e$ with $s \in_R S_{\eta_1}^{\ell}$ and $e \in_R S_{\eta_2}^{k}$) and $z \in_R R_q^k$ with probability $\frac{1}{2}$.
- Required: Determine whether (A, z) is an MLWE instance (so $z = t$) or not.

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Why lattices?

Lattices play two roles in assessing the hardness of MLWE, D-MLWE and MSIS.

1. In 2012, Langlois and Stehlé proved that the *average-case* hardness of MLWE, D-MLWE and MSIS is at least that of the *worst-case quantum* hardness of certain natural lattice problems in certain structured lattices. However, this provable guarantee is highly asymptotic in nature, and so it isn't clear what assurances, if any, it offers about the hardness of MLWE, D-MLWE and MSIS in practice.



https://www.latticechallenge.org/

2. The MLWE, D-MLWE and MSIS problems can be rephrased as lattice problems. These lattice problems are being intensively studied, and the fastest attacks known are used to assess the hardness of MLWE, D-MLWE and MSIS, and thereby justify concrete parameters for Kyber and Dilithium.

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